

Decision Latency and Flow Diagnosis

What decision latency is, why it matters, and the structural patterns that cause it most often. Value stream analysis as a diagnostic tool for knowledge work: how to build a flow diagram, what it reveals, and how it connects to decision protocol design.

What this resource is for

AI accelerates delivery at the individual level. It does not automatically accelerate the decision-making, review, and coordination that surrounds that delivery. The result is a widening gap: teams can produce output faster than ever, but that output still needs to be reviewed, approved, prioritised, and integrated with the work of other teams. In many organisations, the decision-making processes that govern this work have not been updated to match the new tempo. Decision latency — the time between when a decision is needed and when it is made — has become one of the most significant constraints on actual delivery throughput.

In AI-accelerated organisations, the bottleneck to delivery is rarely technical capability. Teams can produce output faster than ever — but that output still needs to be reviewed, approved, prioritised, and coordinated with other teams. The decision-making processes that surround the work are often dramatically slower than the work itself. This resource covers two related diagnostic tools: decision latency analysis, which identifies where decisions are slowing throughput, and value stream mapping, which provides the broader organisational view in which decision bottlenecks become visible.

Decision latency: the invisible tax on speed

Decision latency is the elapsed time between when a decision is needed and when it is made. It is one of the most underappreciated sources of delivery slowdown, because it is often invisible in standard project tracking. A task may be "in progress" for three weeks, but if you examine the timeline carefully, two of those weeks were spent waiting — for approval, for a missing stakeholder to weigh in, for a meeting that finally happened, for clarity on a dependency.

Decision latency compounds in complex organisations. A single delayed decision can block multiple downstream teams simultaneously, and the cost of that delay multiplies across everyone waiting. In an AI-accelerated environment, where the tempo of production has increased but governance structures have not been updated to match, decision latency is frequently the dominant constraint on actual delivery throughput.

Three structural patterns account for most decision latency in practice.

Centralised authority with distributed execution is the most common pattern. Decisions are made at a level of the organisation that is several steps removed from where the work happens. The people doing the work understand the problem deeply and could make a good decision quickly, but they are required to escalate. Each escalation step adds latency, introduces communication loss, and often produces a decision made by someone with less context.

Missing decision owners occur when no one is clearly accountable for a decision. In organisations with highly collaborative or consensus-oriented cultures, this is surprisingly common. The decision gets discussed in meetings, passes through inboxes, and is referenced in documents — but because no one is explicitly responsible for making it, it drifts. Everyone assumes someone else will close it.

Unclear decision criteria occur when people are not sure what counts as a good enough answer. Without clear criteria, reviews become open-ended, stakeholders feel entitled to relitigate settled questions, and decisions never quite reach closure.

Diagnosing decision latency

A decision latency diagnosis examines a recent delivery cycle and asks specific questions about each decision point encountered.

For each significant decision in the period under review, identify: when the need for the decision became apparent; when the decision was actually made; who made it; who was consulted; and whether the outcome was later reversed or relitigated. The gap between the first and second items is your latency figure. The other data points tell you what caused it.

Look for patterns. If most delayed decisions share a common decision-maker, the bottleneck is a person (or a role), and the structural solution is either delegation or a clearer mandate. If most delayed decisions involve cross-team coordination, the bottleneck is a missing forum or protocol. If decisions are frequently relitigated after being made, the bottleneck is in how decisions are documented and communicated.

The goal of the diagnosis is not to assign blame but to produce a structural map of where authority, accountability, and coordination are misaligned with the tempo of delivery.

Value stream mapping as a diagnostic lens

Value stream mapping (VSM) is a lean practice originally developed for manufacturing, adapted widely for knowledge work. A value stream is the complete sequence of activities required to deliver value to a customer — from the moment a need is identified to the moment it is fulfilled. Mapping the value stream means making that sequence visible, including all the waiting, handoff, and rework that happens in between the productive steps.

In knowledge work and software delivery, value stream maps typically reveal that a surprisingly small proportion of total elapsed time is actual productive work. The rest is waiting — for approvals, for dependencies, for capacity, for decisions. This ratio (value-adding time versus total elapsed time) is called the flow efficiency or process cycle efficiency, and it is frequently under 20% even in well-run organisations. In organisations with significant decision latency, it can be under 5%.

How to build a value stream map

A value stream map for a delivery flow typically covers the following elements.

Work stages are the discrete steps that a unit of work passes through from inception to delivery. In a software team, this might include: idea or request intake, refinement, development, code review, testing, deployment, monitoring. Each stage is represented as a box.

Handoffs are the transitions between stages. Each handoff is an opportunity for delay. Note who hands off to whom, and how the handoff is triggered. Many delays hide in handoffs that are assumed to be instantaneous but are actually batched or queued.

Wait times are the elapsed time a unit of work spends sitting between stages, waiting for the next stage to pick it up. These are distinct from the time spent actively working on the item within a stage. Wait times are represented between the stage boxes and are almost always larger than people expect.

Process times are the time spent actively working on the item within each stage.

Decision points are moments where a human judgment is required before the work can proceed. These may be explicit (a formal approval gate) or implicit (a developer waiting for a colleague to clarify a requirement before continuing).

Once the map is built, the diagnostic questions become: where is most of the elapsed time being spent? Are the long waits in work stages or between them? Which decision points have the highest latency? What would need to change to halve the total elapsed time?

Connecting VSM to decision protocol design

Value stream mapping is a powerful setup for decision protocol design because it makes the cost of poor decision-making visible in concrete terms. A decision that takes two weeks to reach closure is not an abstract governance problem — it is a visible delay in the map, with a calculable impact on throughput and customer value.

When participants map their own value streams and identify decision bottlenecks, the conversation about decision protocols becomes practical rather than theoretical. The question shifts from "what is best practice for decision-making?" to "what needs to be true about how we make decisions here in order to remove this specific delay?"

[This is the frame for the decision protocol design work in the Decision-Making and Alignment module.](#)

Further reading

- Womack, J. & Jones, D. — Lean Thinking (source of the value stream concept in manufacturing)
- Reinertsen, D. — The Principles of Product Development Flow (applies flow and queue theory directly to knowledge work)

- Skelton, M. & Pais, M. — Team Topologies (covers the organisational design conditions that determine decision flow)

Online version with links to related resources: <https://resources.simsab.net/coordination-collaboration-ai/decision-latency-and-flow-diagnosis>